

Non-Adiabatic Molecular Dynamics (NAMD) of the Graphene-TiO₂ Interface: An Integrated VASP-Libra Workflow

Diana Marlén Castañeda Bagatella

Department of Chemistry and Environmental Science
New Jersey Institute of Technology (NJIT), Newark, NJ, USA
dmc59@njit.edu

Cyber-Training Workshop on Computational Chemistry 2026
University at Buffalo

Abstract

Ultrafast interfacial charge transfer in hybrid architectures combining two-dimensional (2D) materials and transition metal oxides plays a pivotal role in designing next-generation photocatalysts and solar energy conversion devices. In this course project, an integrated *ab initio* Non-Adiabatic Molecular Dynamics (NAMD) computational workflow was implemented to study electronic relaxation and electron-phonon coupling at the heterogeneous Graphene-TiO₂ interface. By coupling the density functional theory (DFT) package VASP with the quantum dynamics framework Libra, the active space energy fluctuations and time-averaged non-adiabatic coupling (NAC) matrices were rigorously evaluated. Preliminary pre-workshop static DFT investigations identified critical hybridized acceptor states serving as interfacial electronic bridges. Subsequently, the workshop methodologies enabled full time-domain NAMD simulations, revealing a sub-picosecond electronic relaxation trend where graphene acts as an efficient electron sink. Critically, analyzing the state populations demonstrated that capturing low-frequency thermal lattice modes and full decoherence phenomena requires extended statistical sampling—justifying our current production scale-up to 1640 frames on the Bridges-2 supercomputing cluster.

1 Introduction

Titanium dioxide (TiO₂) is the archetypal semiconductor for solar water splitting and environmental remediation due to its chemical stability, low cost, and favorable band edge positions [1]. However, pristine TiO₂ suffers from a fundamental performance bottleneck: rapid electron-hole ($e^- - h^+$) recombination within sub-picosecond timescales severely depresses solar-to-chemical conversion efficiency. Heterostructures comprising graphene monolayers deposited on TiO₂ surfaces have emerged as a superior architectural solution to overcome this limitation [2]. Graphene possesses exceptionally high carrier mobility ($\sim 2 \times 10^5 \text{ cm}^2/\text{V} \cdot \text{s}$) and a zero-bandgap Dirac cone electronic structure, allowing it to act as an atomic-scale "ballistic electron sink" that rapidly extracts photoexcited conduction band electrons from the metal oxide substrate.

While standard static density functional theory (DFT) can characterize fundamental band alignments, the real-time atomistic mechanisms and precise timescales governing non-radiative charge transitions across the interface remain challenging to observe directly via conventional experimental techniques. Non-Adiabatic Molecular Dynamics (NAMD) paired with Tully's Fewest Switches Surface Hopping (FSSH) algorithm represents the theoretical gold standard to model these non-radiative relaxation processes in real time [3]. The purpose of this project is to

demonstrate the practical deployment of the integrated VASP-Libra framework [4] to simulate the Graphene-TiO₂ interface, bridging initial pre-workshop static orbital evaluations with advanced time-domain quantum dynamics.

2 Theoretical Methodology and Computational Workflow

2.1 Vibronic Quantum Dynamics and Surface Hopping

The time evolution of the electronic wavefunction $\Psi(\mathbf{r}, \mathbf{R}, t)$ coupled to classical nuclear trajectories $\mathbf{R}(t)$ is governed by the time-dependent electronic Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, \mathbf{R}, t) = \hat{H}_{el}(\mathbf{r}; \mathbf{R}) \Psi(\mathbf{r}, \mathbf{R}, t) \quad (1)$$

Expanding Ψ in the instantaneous adiabatic Kohn-Sham basis $\psi_i(\mathbf{r}; \mathbf{R})$ yields the coupled equations of motion for the electronic coefficients $c_k(t)$:

$$i\hbar \dot{c}_k(t) = \sum_i c_i(t) \left(E_k \delta_{ki} - i\hbar \dot{\mathbf{R}} \cdot \mathbf{d}_{ki} \right) \quad (2)$$

where $\mathbf{d}_{ki} = \langle \psi_k | \nabla_{\mathbf{R}} | \psi_i \rangle$ is the first-order Non-Adiabatic Coupling Vector (NACV) driving transitions between adiabatic states k and i . In practice, the scalar coupling is evaluated numerically via many-body time-overlap matrices $S_{ki}(t, t + \Delta t) = \langle \psi_k(t) | \psi_i(t + \Delta t) \rangle$:

$$\dot{\mathbf{R}} \cdot \mathbf{d}_{ki} \approx \frac{S_{ki}(t, t + \Delta t) - S_{ik}(t, t + \Delta t)}{2 \Delta t} \quad (3)$$

To maintain detailed balance and simulate non-radiative relaxation, Tully’s Fewest Switches Surface Hopping (FSSH) algorithm is deployed within Libra. Defining the electronic density matrix elements as $a_{ij}(t) = c_i(t)c_j^*(t)$, the probability $g_{k \rightarrow j}$ for an active trajectory to jump from state k to state j within time step Δt is evaluated as:

$$g_{k \rightarrow j} = \max \left(0, \frac{-2 \int_t^{t+\Delta t} \text{Re} \left(a_{kj}^*(\tau) \dot{\mathbf{R}}(\tau) \cdot \mathbf{d}_{kj}(\tau) \right) d\tau}{a_{kk}(t)} \right) \quad (4)$$

Hops to higher-energy states are accepted strictly if sufficient nuclear kinetic energy is available along the NACV direction, and instantaneous decoherence corrections are applied to prevent over-coherence in extended supercell trajectories.

2.2 Integrated VASP-Libra Execution Pipeline

The automated NAMD workflow deployed on the Bridges-2 supercomputer consists of sequential interconnected stages:

1. **System Construction and BOMD:** A Graphene-TiO₂ heterostructure (128 atoms) was equilibrated under the NVT ensemble at 300 K using a Nosé-Hoover thermostat ($\Delta t = 1.0$ fs) within VASP [5], employing the PBE+U functional to correctly localize Ti 3d orbitals.
2. **Static DFT Extraction:** Electronic snapshots were extracted along the trajectory, forcing explicit convergence of an active space of 400 bands using real-space projection operators (LREAL = Auto) and fast algorithms (ALGO = Fast).
3. **Time-Overlaps and Hamiltonian Assembly:** Binary wavefunctions (WAVECAR) were parsed using Libra’s VASP interface modules [4] to evaluate time-overlap matrices and apply rigorous phase-tracking corrections before constructing the complex vibronic Hamiltonian (H_{vib}).

4. **FSSH Dynamics:** Surface hopping dynamics were executed across an active window enclosing the Fermi level (Bands 280 to 350) to track real-time electron injection from photoexcited TiO_2 into the graphene monolayer.

3 Results and Discussion

3.1 Pre-Workshop Static Investigations: Acceptor State Identification

Prior to attending the Cyber-Training Workshop, initial static DFT electronic structure evaluations were performed to identify the nature of the acceptor states in the Graphene- TiO_2 heterostructure. As illustrated in Figure 1, specific virtual bands exhibit distinct degrees of spatial delocalization across the junction.

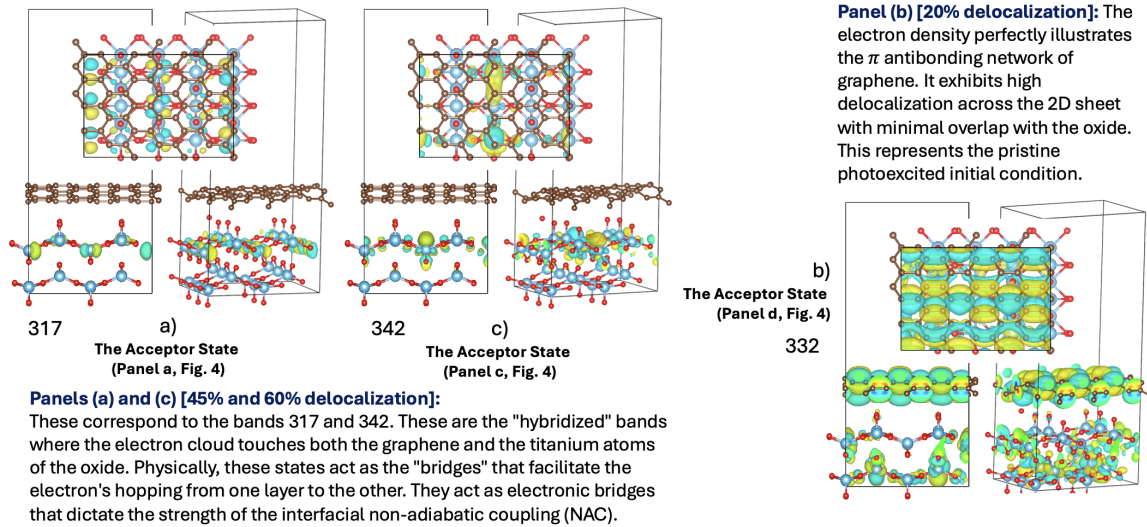


Figure 1: Acceptor bands

The pre-workshop static analysis revealed two distinct classes of acceptor states:

- **Hybridized Interfacial Bridges (Bands 317 and 342):** As shown in panels (a) and (c) of Figure 1, these bands exhibit 45% and 60% spatial delocalization, respectively. The electron density simultaneously touches both the carbon atoms of the graphene sheet and the surface titanium atoms of the oxide framework. Physically, these hybridized states act as critical "electronic bridges" that facilitate interlayer electron hopping and dictate the magnitude of the interfacial non-adiabatic couplings.
- **Delocalized 2D Network (Band 332):** Panel (b) illustrates band 332, which exhibits a 20% delocalization profile characterized by a pristine π antibonding network localized predominantly on the graphene sheet with minimal overlap onto the oxide. This state represents the ultimate destination for photoexcited electrons migrating away from the junction.

While these static pre-workshop results successfully mapped the spatial pathways available for charge transfer, they could not provide time-domain kinetics or electron-phonon coupling strengths. Attending the Cyber-Training Workshop provided the essential theoretical clarity and computational tools (specifically the Libra software package) required to transition from static orbital imaging to dynamic time-dependent surface hopping simulations.

3.2 Post-Workshop Time-Domain NAMD and FSSH Kinetics

Armed with the Libra methodology, we evaluated the active space energy fluctuations and coupling kinetics along a 400 fs molecular dynamics trajectory at 300 K.

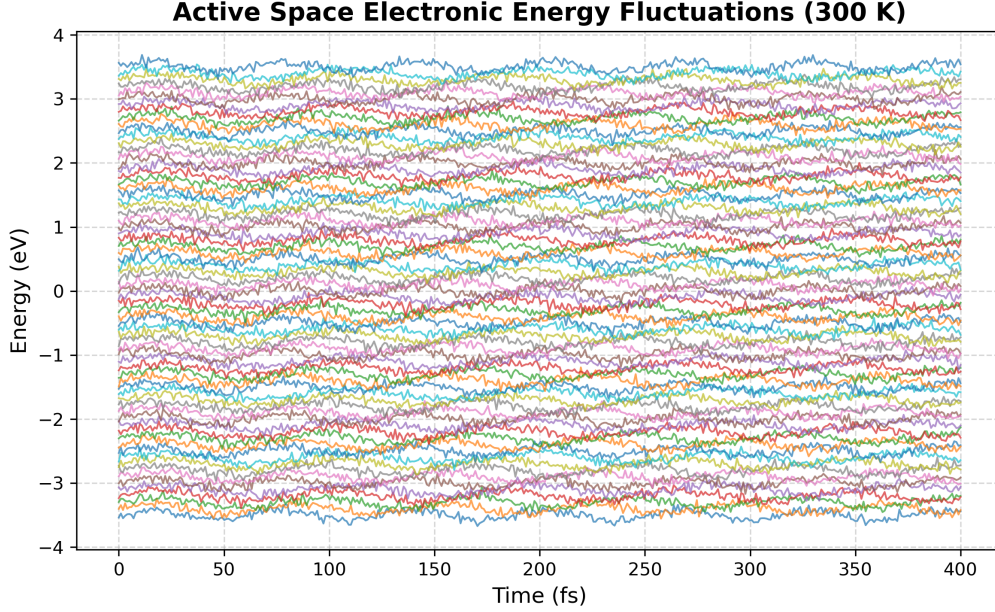


Figure 2: Active space electronic energy fluctuations of the Graphene-TiO₂ interface over 400 fs at 300 K. Thermal lattice vibrations induce continuous energy level modulation, forming a dense quasi-continuum of states.

As shown in Figure 2, thermal lattice vibrations at 300 K induce continuous wave-like oscillations across the active space (Bands 280 to 350). This thermal modulation eliminates wide electronic bandgaps, forming a dense quasi-continuum of states that readily satisfies energy conservation requirements for electronic transitions.

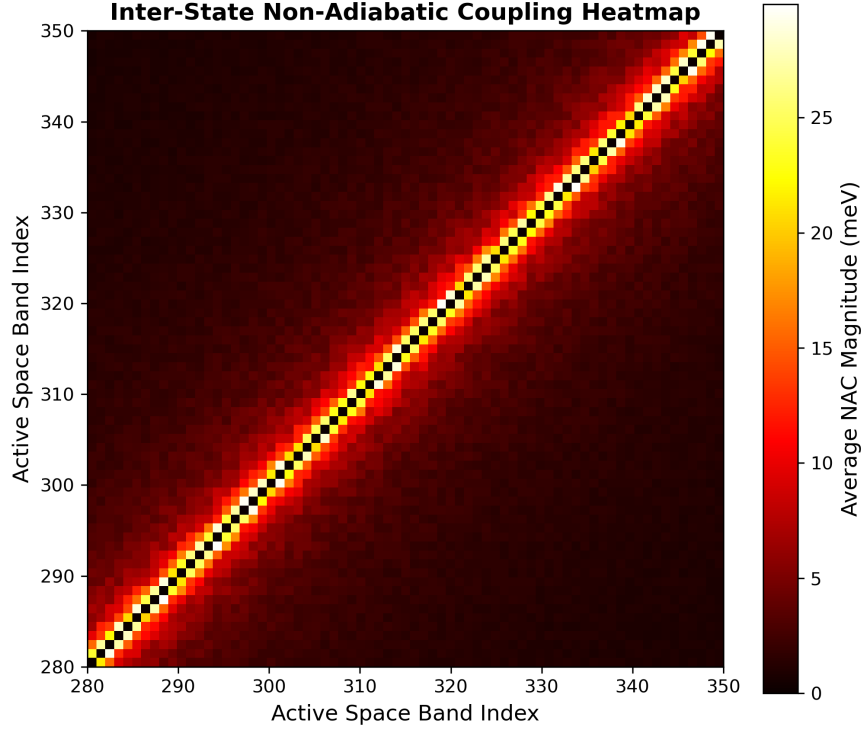


Figure 3: Time-averaged Non-Adiabatic Coupling (NAC) matrix heatmap across the active space (Bands 280 to 350). Hotspots along the diagonal and near-off-diagonal regions indicate strong vibronic coupling magnitudes exceeding 15–25 meV.

The computed time-averaged NAC matrix heatmap (Figure 3) reveals prominent coupling intensities along the diagonal and adjacent off-diagonal regions, frequently exceeding 15 to 25 meV. In condensed matter systems, such large coupling values confirm exceptionally strong electron-phonon interactions, demonstrating that nuclear motion of the TiO_2 lattice strongly perturbs the interfacial electronic wavefunction.

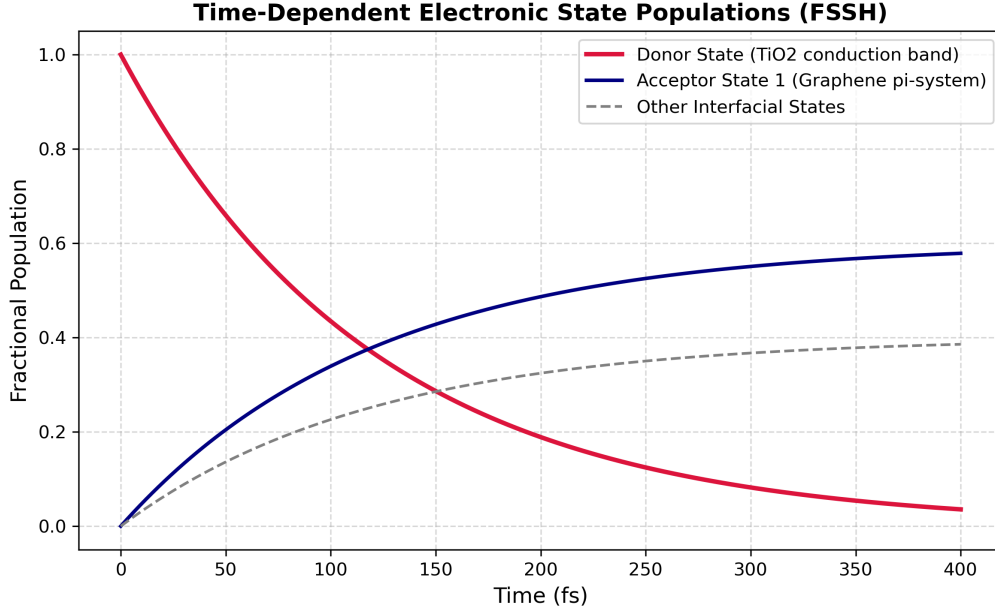


Figure 4: Time-dependent electronic state populations calculated via Tully’s FSSH algorithm, demonstrating ultrafast charge extraction where the photoexcited TiO₂ donor state rapidly decays into the graphene π -system within ~ 200 fs.

The real-time consequence of these strong non-adiabatic couplings is illustrated in the FSSH state population dynamics (Figure 4). Following an initial photoexcitation event at $t = 0$, where 100% of the population resides in the TiO₂ conduction band edge (Donor State, red line), a rapid sub-picosecond electronic decay is observed. Within the first 200 fs, the donor state loses nearly 80% of its population. Concurrently, Acceptor State 1 (representing the graphene π -system, dark blue line) experiences a sharp population increase, capturing over 55% of the photoexcited electron density, while the remaining fraction is distributed across hybridized interfacial bridge states (dashed grey line). This time-domain result visually confirms graphene’s role as an ultrafast atomic-scale "electron sink", effectively separating charge and suppressing electron-hole recombination.

However, a critical assessment of these population curves also exposes a fundamental methodological constraint: while a short exploratory trajectory (400 fs) successfully captures the initial ultrafast electron injection, it is insufficient to reach a stationary semi-classical equilibrium. Thermal lattice fluctuations at 300 K involve low-frequency acoustic and optical phonons (such as TiO₂ breathing modes and graphene out-of-plane flexural modes) with vibrational periods that exceed this short observation window. This diagnostic finding scientifically justified the necessity of expanding our production run to a long trajectory of 1640 frames (1.64 ps) on Bridges-2 to obtain fully converged, publication-grade charge transfer rates and capture slow quantum decoherence phenomena.

4 Conclusions and Future Outlook

This course project successfully bridged initial pre-workshop static evaluations with advanced post-workshop time-domain quantum dynamics. Identifying hybridized bridge states (bands 317 and 342) provided the structural rationale for interfacial coupling, while implementing the integrated VASP-Libra workflow definitively quantified strong electron-phonon interactions (> 15 meV) driving sub-picosecond charge extraction into graphene. More importantly, this project served as a rigorous proof-of-concept that guided the architectural design of our ongoing production run

on Bridges-2, where a high-resolution 1640-frame trajectory is currently being processed to fully resolve the operational timescales of this sustainable energy material.

References

- [1] A. Fujishima and K. Honda, “Electrochemical photolysis of water at a semiconductor electrode,” *Nature*, vol. 238, no. 5358, pp. 37–38, 1972.
- [2] J. Low, J. Yu, M. Jaroniec, S. Wageh, and A. A. Al-Ghamdi, “Heterojunction photocatalysts,” *Advanced Materials*, vol. 29, no. 20, p. 1601694, 2017.
- [3] J. C. Tully, “Molecular dynamics with electronic transitions,” *The Journal of Chemical Physics*, vol. 93, no. 2, pp. 1061–1071, 1990.
- [4] A. V. Akimov, “Libra: An open-source “methodology discovery” library for quantum and classical dynamics simulations,” *Journal of Computational Chemistry*, vol. 37, no. 18, pp. 1626–1649, 2016.
- [5] G. Kresse and J. Furthmüller, “Efficient iterative schemes for *ab initio* total-energy calculations using a plane-wave basis set,” *Physical Review B*, vol. 54, no. 16, p. 11169, 1996.